

Infosys Springboard Virtual Internship 7.0

Completion Report

Team Details

Batch Number : 1

Start date : 29th June 2026

Name: SUJAN M A

Internship Duration: 8 Weeks

1. Project Title

CivicPulse – Smart Governance Platform.

2. Project Objective

The main objective of the **CivicPulse** internship project was to develop a centralized digital platform for managing civic services and administrative operations while providing citizens with an easy way to report issues, apply for documents, and stay informed through announcements. The system aimed to provide an efficient communication channel between citizens, field officers, and administrators, enabling issues and applications to be submitted, assigned, monitored, and resolved in a structured manner.

The project focused on improving the transparency and efficiency of civic administration by providing role-based access, real-time status tracking, officer assignment with an accept workflow, document application processing with file delivery, and centralized announcements. Citizens can report problems, apply for documents, and track their progress, while officers can accept and manage assigned complaints and update their resolution status. Administrators can oversee the overall system, manage officer accounts, review applications, and publish announcements.

From an organizational perspective, CivicPulse demonstrates how a modern web-based system can digitize and streamline civic service delivery. It reduces manual effort, improves coordination between different stakeholders, and provides a scalable foundation for better administrative operations and citizen service delivery.

3. Project description in detail

CivicPulse – Smart Governance Platform is a web-based civic administration platform that centralizes citizen-related services — reporting and managing public complaints, processing government document applications, assigning complaints to officers, and publishing public announcements.

The project follows a full-stack architecture where a React.js frontend communicates with a Java Spring Boot backend through REST APIs. Role-based access for Citizen, Officer, and Admin ensures each user can only perform operations relevant to their responsibilities.

Technology Used

- **Frontend:** React.js, Vite, HTML, CSS, JavaScript
- **Backend:** Java, Spring Boot
- **Database:** MySQL
- **Security:** Spring Security, JWT
- **File Handling:** Multipart file upload/download for document applications
- **API Testing:** Postman
- **Version Control:** Git and GitHub

Development Approach

The backend was built modularly, with REST APIs for user management, complaint reporting/assignment/status tracking, document applications, announcements, and admin operations. The frontend consumes these APIs to provide role-specific interfaces. Authentication used Spring Security and JWT; MySQL handled data storage; Postman was used for API testing throughout development.

Real-World Impact

CivicPulse shows how civic services can move from manual processes to a centralized digital system — letting citizens submit complaints and applications online while giving officers and admins the tools to manage, process, and track them efficiently, improving transparency and coordination.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand project requirements and study existing civic administration systems	Analyzed project requirements and finalized the major modules of CivicPulse
Week 2	Design system architecture and database	Designed the application architecture and MySQL database structure
Week 3	Develop backend structure and REST APIs	Created Spring Boot project and implemented core REST APIs
Week 4	Implement authentication and user roles	Implemented JWT authentication, Spring Security, and role-based access for Citizen, Officer, and Admin
Week 5	Develop complaint and civic service management modules	Implemented complaint reporting, officer assignment, complaint status management, and document application processing
Week 6	Develop React frontend and integrate APIs	Built frontend screens and integrated them with the Spring Boot REST APIs
Week 7	Implement announcements, file handling, and improve system functionality	Implemented public announcements, document upload/download functionality, and improved application features
Week 8	Testing, debugging, and final improvements	Tested the complete application, fixed frontend and backend issues, validated APIs using Postman, and finalized the project

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project requirements were discussed and the development plan was finalized.	Week 1
Prototype/First Draft	Initial CivicPulse prototype with basic frontend and backend functionality was completed.	Week 3
Mid-Term Review	Core modules, authentication, issue management, and database integration were reviewed.	Week 5
Final Submission	Complete system with frontend, backend, database, security, and caching was finalized.	Week 8
Presentation	Project features, implementation, and results were presented.	Week 8

5b. Project execution details

The CivicPulse – Smart Governance Platform was executed through requirement analysis, system design, development, testing, and finalization phases.

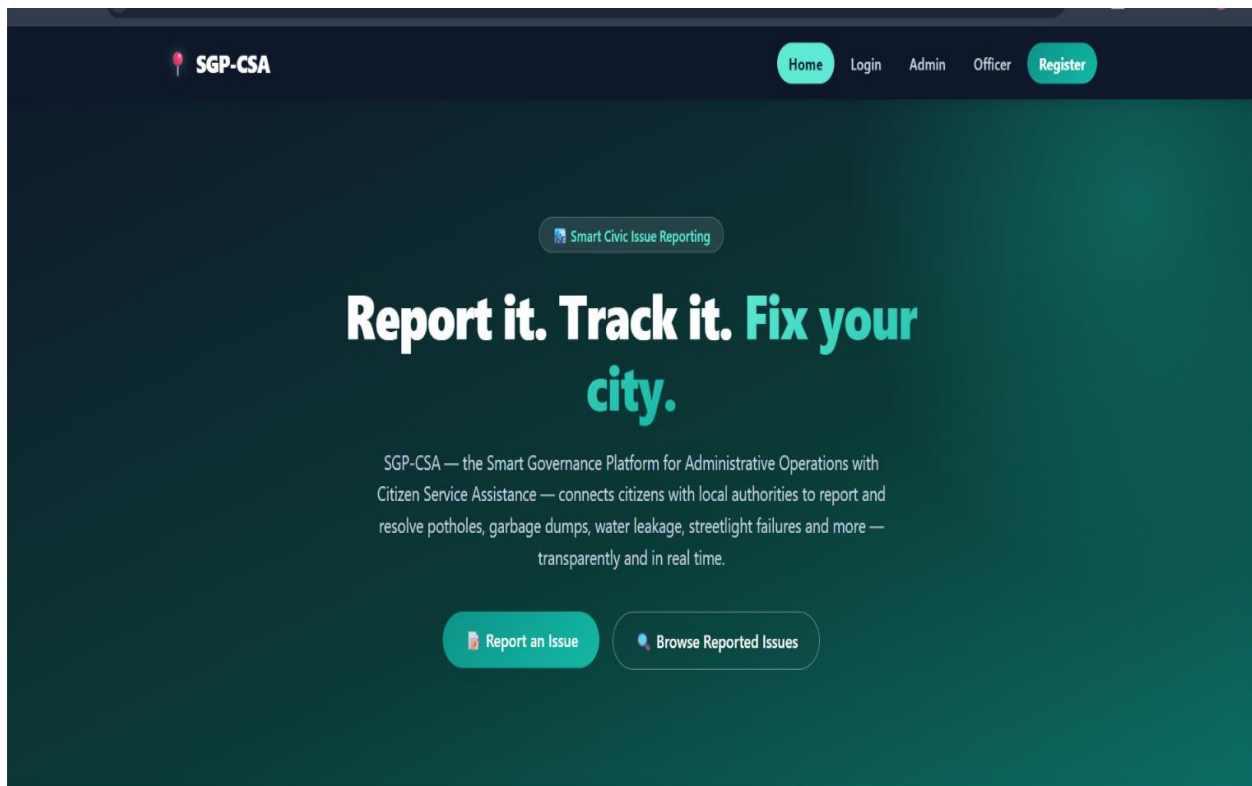
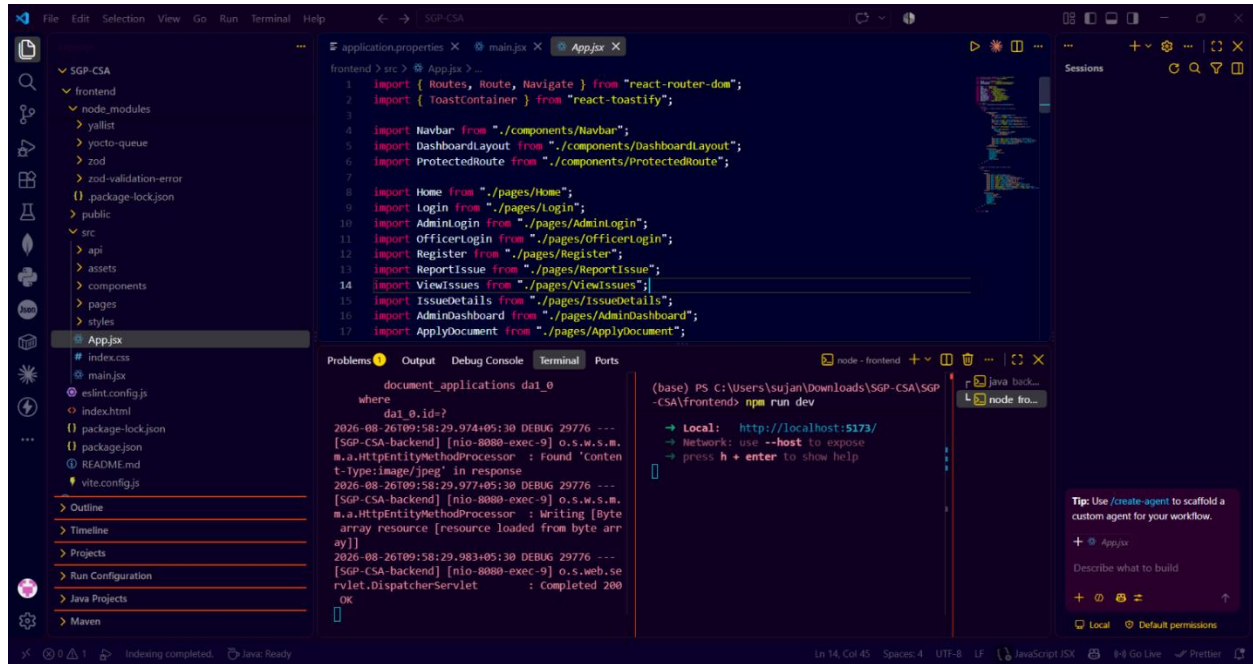
The backend was developed using Java and Spring Boot with REST APIs, while MySQL was used for database management.

The frontend was developed using React.js and Vite and integrated with the backend through REST APIs for Citizen, Officer, and Admin roles.

Spring Security, JWT authentication, and multipart file handling were implemented for secure access and document management, with Postman used for API testing.

Finally, the complete system was tested and debugged to resolve integration, authentication, database, and API issues before final submission.

6.Snapshots / Screenshots



SGP-CSA

HomeLoginAdminOfficerRegister

Welcome Back

Login to report and track civic issues.

Email

Password

Login

Don't have an account? [Register](#)

Are you an admin? [Admin login](#)

Are you an officer? [Officer login](#)

SGP-CSA | Smart Governance | X +

Ask Gemini

localhost:5173/apply

SGP-CSA

Home

Report Issue

View Issues

Apply for Documents

Announcements

S Sujan M A
Citizen

Logout

Welcome back, Sujan 🌟

Apply for a Government Document

Apply for Aadhar Card, Birth Certificate, or other civic documents online.

Document Type

Aadhar Card

Applicant Full Name

As per official records

Contact Number

10-digit mobile number

Address

Full residential address

Remarks (optional)

Any additional details...

Submit Application

My Applications

Aadhar Card

Sujan M A · 91235487333

Approved

✓ Your document has been approved and sent to you. Please check your registered contact / email, or visit the municipal office to collect it.

Applied: 25/08/2026

Admin: Your document is ready. Please visit the municipal office with your ID to collect it.

Birth Certificate

Sujan M A · 9123548733

Approved

✓ Your document has been approved and sent to you. Please check your registered contact / email, or visit the municipal office to collect it.

Applied: 26/08/2026

Admin: Your document is ready. Download it below.

Download Document

SGP-CSA

Home

Report Issue

View Issues

Apply for Documents

Announcements

Sujan M A
Citizen

Logout

Welcome back, Sujan 🍌



Report a Civic Issue

Tell us what's wrong and where — we'll route it to the right team.

Issue Title

e.g. Large pothole on Main Street

Description

Describe the issue in detail...

Location

e.g. 5th Cross, Indiranagar

Use My Location

Category

Select Category

Report Issue

SGP-CSA

Home

Report Issue

View Issues

Apply for Documents

Announcements

Sujan M A
Citizen

Logout

Welcome back, Sujan 🍌



Latest Announcements

PLEASE PAY THE CURRENT BILL BEFORE 27-08-26

PAY AS SOON AS POSSIBLE

 Due: 27/08/2026 Posted: 26/08/2026

SGP-CSA

Home

Report Issue

View Issues

Apply for Documents

Announcements

Dashboard

Abdul Admin

Administrator

Logout

Welcome back, Abdul 🌟

Admin Dashboard

Total Issues

1

Pending

0

In Progress

1

Resolved

0

Quick Actions

Report Issue

View All Issues

Manage Officers

Officer Name

admin@civulpulse.com

+ Add Officer

Default Officer (officer@civulpulse.com)

rahul (rahul@civulpulse.com)

SGP-CSA

Home

Report Issue

View Issues

Apply for Documents

Announcements

Dashboard

Abdul Admin

Administrator

Logout

Officer Name

admin@civulpulse.com

+ Add Officer

Default Officer (officer@civulpulse.com)

rahul (rahul@civulpulse.com)

Document Applications

APPLICANT	DOCUMENT	CONTACT	STATUS	ACTION
Sujan M A	Aadhar Card	91235487333	Approved	Update status...
Sujan M A	Birth Certificate	91235487333	Approved	Update status...
Sujan M A	Birth Certificate	91235487333	Pending	Update status...

Search issues...

All

Recent Issues

TITLE	CATEGORY	LOCATION	STATUS	ASSIGNED OFFICER
tree hazard	Road	link	In Progress	Unassigned

pg. 8

7.Challenges Faced

- **Role-Based Authentication:** Implementing three separate login flows (Citizen, Officer, Admin) with JWT-based authentication required careful configuration of Spring Security to ensure each role could only access its permitted routes and APIs. This was resolved by structuring dedicated login endpoints and validating role claims in the JWT filter.
- **Officer Assignment Workflow:** Coordinating the flow where an Admin assigns an issue to an Officer, the Officer accepts it, and the status auto-updates was tricky to get consistent. This was solved by clearly defining issue status states and updating them at each stage of the workflow in the backend service layer.
- **Document Application Processing:** Building the flow for citizens to apply for documents (Aadhar Card, Birth Certificate, etc.) and have Admins approve/reject with a custom response message required careful entity design linking applications to users and tracking their status.
- **Location Capture & Maps Integration:** Capturing device GPS location for reported issues and linking it to Google Maps for admins/officers to view took some trial and error with browser geolocation APIs and coordinate handling.
- **Frontend-Backend Integration:** Connecting the React (Vite) frontend to the Spring Boot REST APIs using Axios, especially for protected routes, required debugging token handling and CORS configuration.
- **Database Design:** Structuring MySQL relationships between Users, Issues, Document Applications, and Announcements to support role-based access and status tracking took iteration to get right using Spring Data JPA.

8. Learnings & Skills Acquired

During the internship, I gained hands-on experience in full-stack development by building **CivicPulse – Smart Governance Platform**.

- Strengthened skills in **Java, Spring Boot, and Spring Data JPA** for backend API development.
- Learned to design and implement **role-based authentication** using Spring Security and JWT across multiple user types.
- Built and consumed **REST APIs** connecting a React (Vite) frontend with Axios and React Router.
- Learned to structure a **relational database in MySQL** to support multi-role workflows and status tracking.
- Gained experience implementing **protected routes**, toast notifications, and a responsive dashboard UI.
- Improved **debugging and testing** skills while resolving integration issues between frontend and backend.
- Developed better **project structuring, version control (Git), and problem-solving** skills while building a real-world civic-services application end to end.

9. Testimonials from team

The internship gave us valuable practical experience in building a complete, real-world civic services platform. Working on **CivicPulse** helped us understand how authentication, role management, and multi-user workflows come together in a production-style application.

Collaborating on the assignment and approval workflows — between citizens, officers, and admins — improved our teamwork and communication, and gave us confidence in applying our technical skills to solve real administrative problems.

10. Conclusion

This internship provided valuable hands-on experience in full-stack application development through the **CivicPulse – Smart Governance Platform** project. It helped me apply academic knowledge to a real-world problem involving multi-role authentication, civic issue management, and government document application workflows.

Building the platform improved my skills in **Java, Spring Boot, React, MySQL, and REST API development**, while also strengthening my **problem-solving, teamwork, and communication** abilities. Overall, the internship reinforced my technical foundation and aligned well with my career goal of becoming a **software/full-stack developer**.

11. Acknowledgements

I sincerely thank the **organization** for providing me with the opportunity to work on the **CivicPulse** project. I am grateful to my **mentor** for their guidance, feedback, and support throughout the internship.

I also thank my **team members** for their cooperation, support, and valuable contributions. This internship was a great learning experience that helped me grow both technically and professionally.